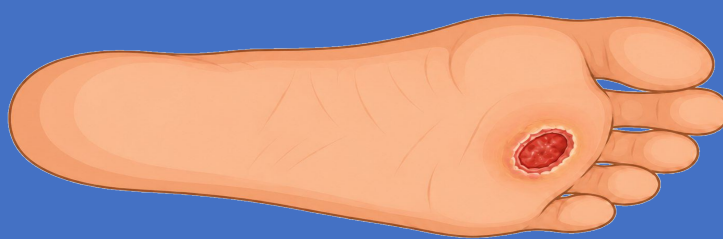


Background

- Over **537 million people worldwide** live with diabetes, and it is expected to reach **643 million by 2030 [1]**
- Diabetes can cause **peripheral neuropathy**, preventing patients from detecting pain and pressure
- 19% to 34% of people** with diabetes develop a foot ulcer during their lifetime



Objectives

- Develop a compact system to assist clinician-guided offloading of high-pressure diabetic foot regions
- Automatically detect high-pressure regions associated with ulcer risk
- Relieve pressure off the foot ulcer
- Build a platform that can provide a gait oriented response

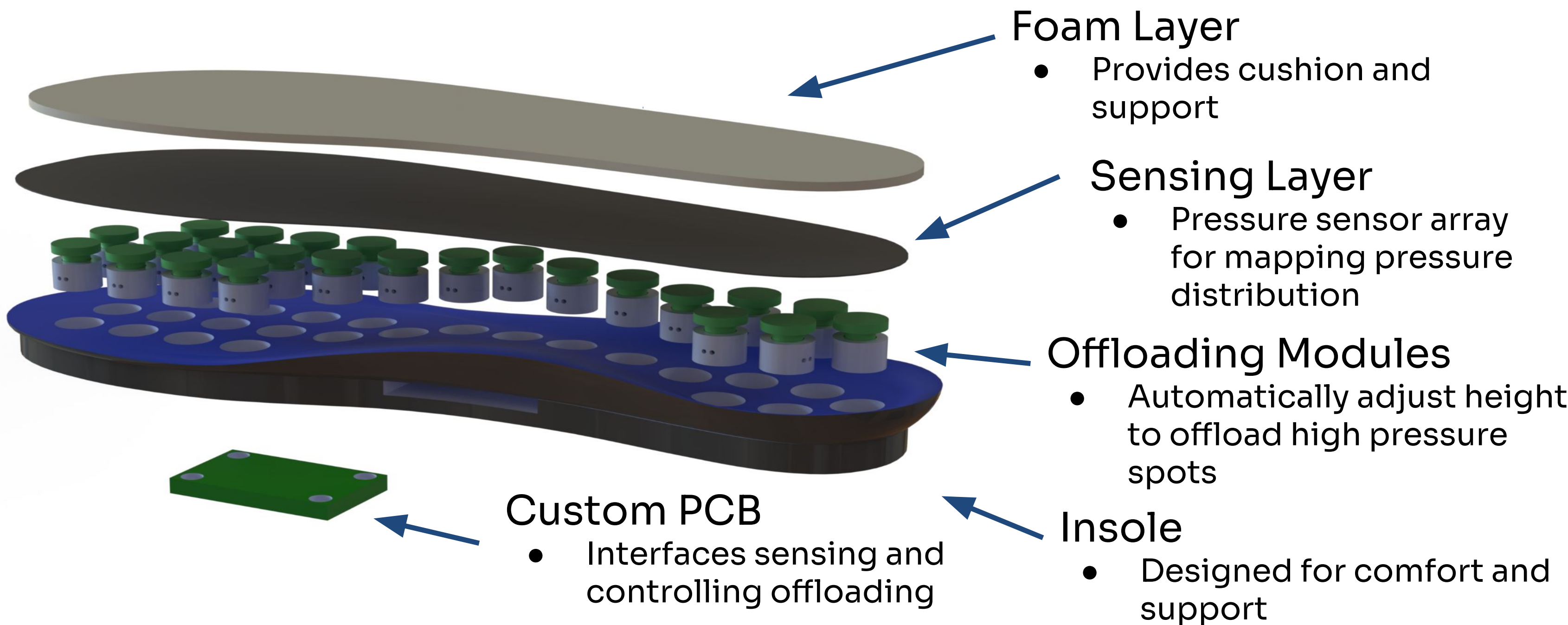


Requirements

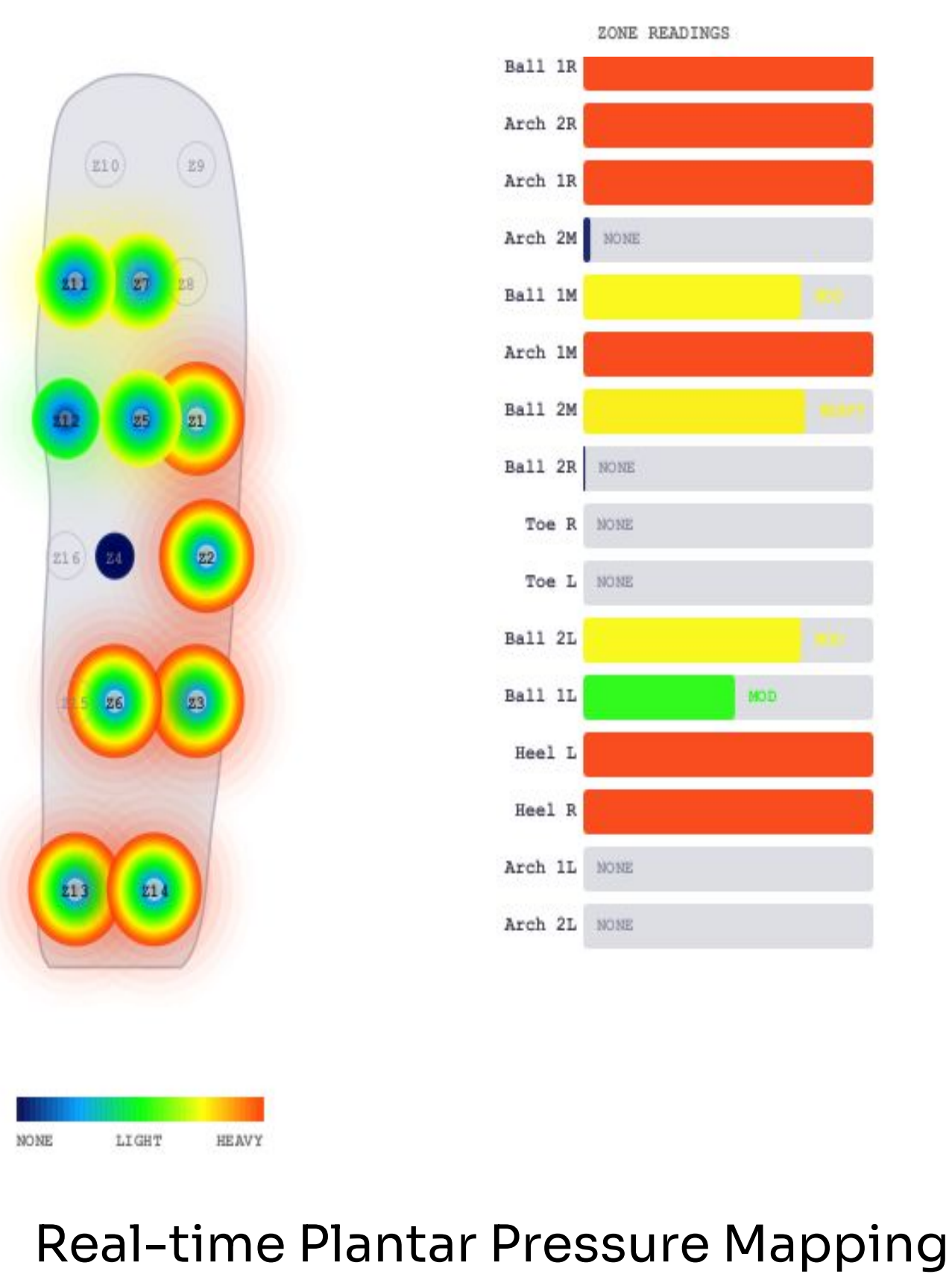
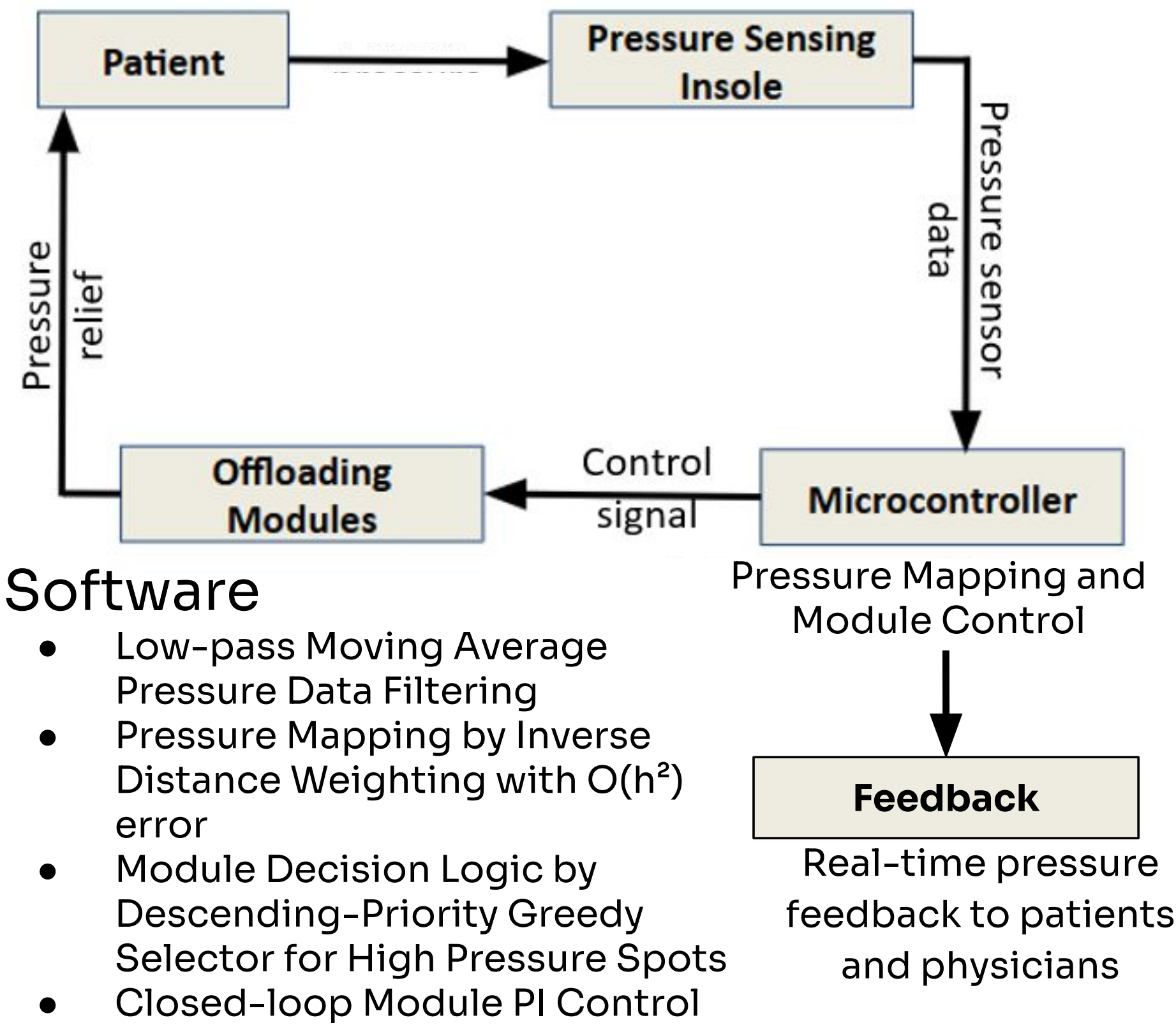
- ✓ Offload pressure by 30-50%
- ✓ Withstand 800-1200 kPa peak plantar pressure
- ✓ Fit within a 20 mm height constraint
- ✓ Identify heel strike, midstance, and toe liftoff
- ✓ Transition between offloading and rigid support

Special Thanks: Dr. Alyssa Taylor, Dr. Sean Collignon, Nagarjun Bhat, Greg Hainline, DIB Staff

Hardware Design



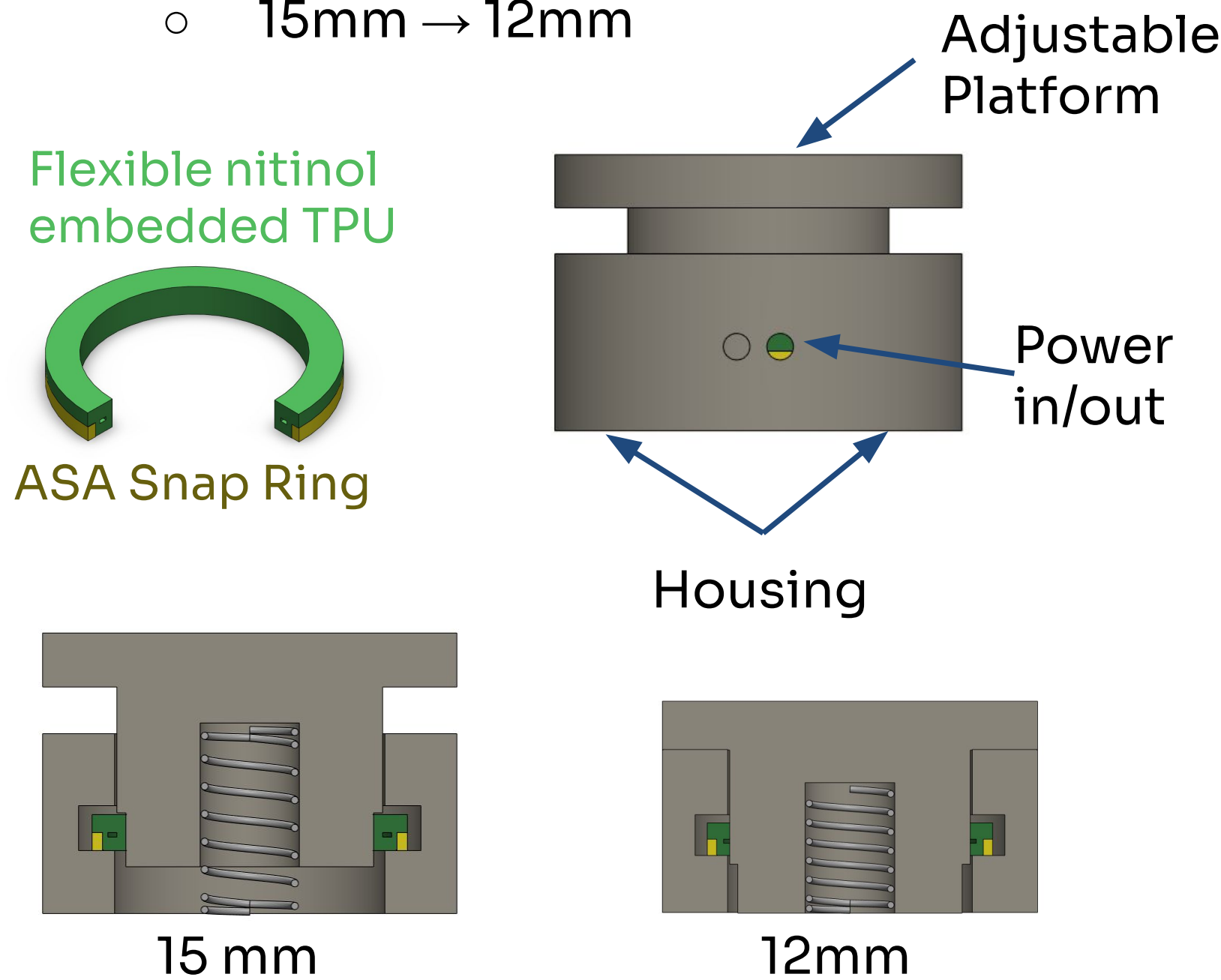
Sensing & Control Architecture



Offloading Modules Design

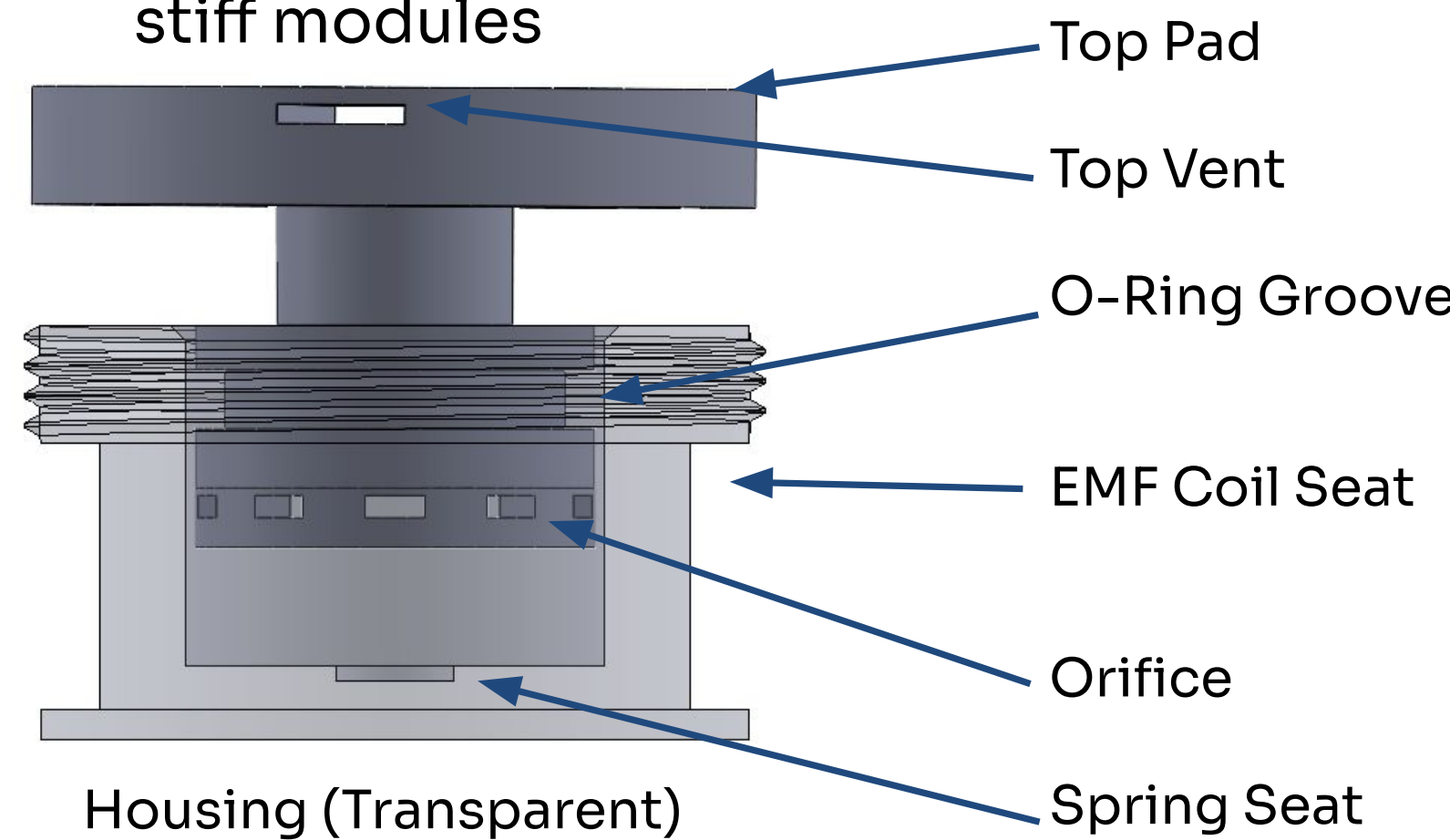
Nitinol Lock Ring Module

- Lock ring expands when power is applied
- Allows platform to fall into spring supported offloading state
 - 15mm → 12mm

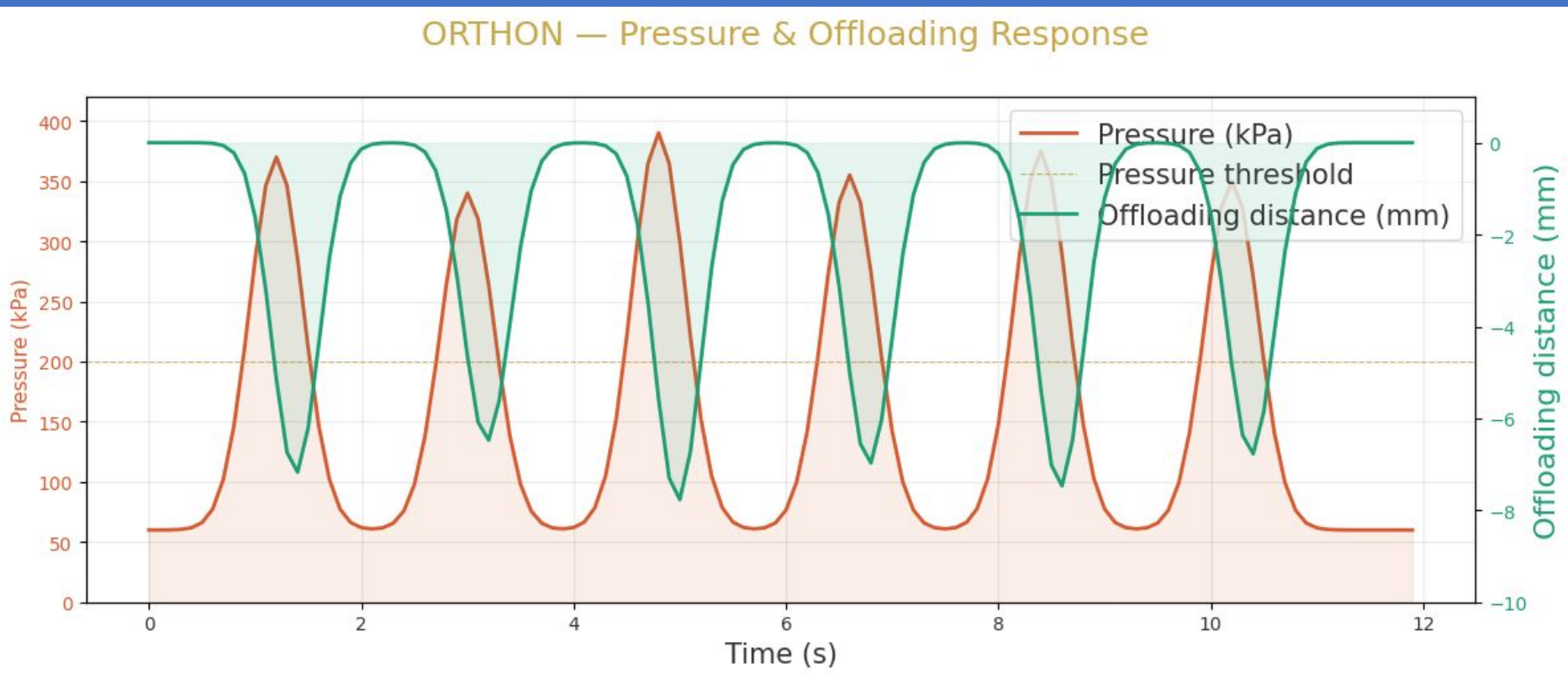


MRF (Magnetorheological Fluid) Module

- Magnetic Field strength increases MRF flow resistance through piston orifice allowing selectivity of soft and stiff states
- Offloading occurs by users stepping on the modules; Soft modules transfer forces to stiff modules

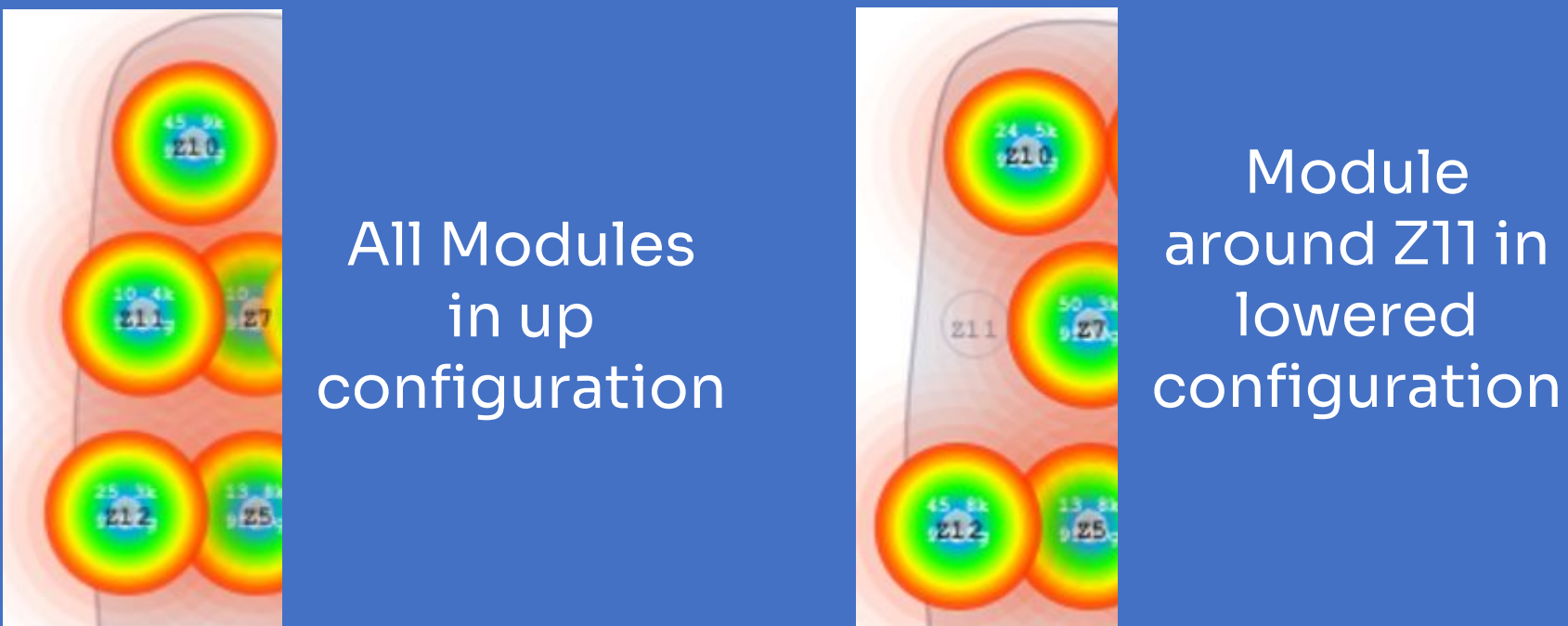


Analysis & Testing Results



Simulated gait offloading response
Each excess pressure spike is followed by a corresponding rise in offloading distance, the actuator drops to redistribute load

Lowering Module Test Results:



Lowered modules correspond to decreased localized pressure. Hence, the localized loading is directly controlled by module height, and validates offloading strategy

Conclusions

Orthon **successfully demonstrates real-time closed pressure offloading** at both the forefoot and heel across recorded tests

Results **validate concept viability**

Future Steps

- Transition to a compact, fully wearable device
- Develop Virtual Twin software
- Conduct trials with diabetes and post-surgical cohorts
- Evaluate regulatory pathway toward Class II device classification